Department of Computer Science

CS5705 Ethics and Governance of Digital Systems

Postgraduate Study Guide for 2025/26

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MODULE DETAILS

Module Leader	Dr Steven Sam	
Department	Computer Science	
Credits	15 - This is assessed in the CS5805 block (unless on DTSS)	
Other staff	Dr Antonios Kaniadakis Dr Annette Payne Dr Federico Castagna	
Contact and private study time	Lectures Seminars Exam Preparation General Study Total	18 hours 9 hours 70 hours 53 hours 150 hours
Assessment	Method Exam	Single Assessment (CS5805) 100%

ACCESS TO SUPPORT MATERIAL AND ADDITIONAL INFORMATION

Most teaching, learning and support material is provided electronically via the University's Brightspace system. Note that the details provided in this study guide are based on the formal module syllabus for this module which sets out the agreed content, learning outcomes, assessment and teaching methods. Module syllabus and scheme of studies documents for your programme of study can be found by on the University's Quality Assurance web pages.

INTRODUCTION/AIMS/BACKGROUND

The aim of this module is to:

> *Develop a critical understanding of topics related to the handling and governance of digital information in contemporary systems contexts*

Digital information systems are both technical and social in nature. Therefore, when developing or using such systems, decision makers and users should understand the mutual influences between the technical aspects of the systems and the structure, functions and

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politics of organisations and other social groups. Similarly, system developers need an understanding of management objectives and decision-making and the impact that digital information systems will have on the well-being of individuals, organisations and society.

Topics which address the module aim include: the way that digital technologies and networked and intelligent systems are designed and used; the motivations for their adoption; the substantive issues arising; and approaches to their regulation and governance. Examples from the public and private sectors will be used to illustrate these developments.

LEARNING OUTCOMES

Whatever module or programme of study you are studying for at Brunel University London, there are learning outcomes (LO) that you must meet/achieve in order to be awarded the credits which comprise the module and programme of study. In order to get a pass grade (C- or above) in this module, you must meet these learning outcomes below, that is, you must demonstrate ability to:

LO1: Analyse and critically appraise relevant academic and governance sources related to the handling and governance of digital information.

LO2: Evaluate the legal, social and ethical concerns and consequences related to the use of digital systems in practice.

LO3: Reflect critically on the wider organisational and societal implications of digital technologies and justify appropriate governance responses.

METHOD OF TEACHING

The typical methods used to help you to achieve the learning outcomes on this module are:

- Content delivery sessions that concentrate on providing you with frameworks for (a) understanding domain knowledge, (b) understanding activities, and (c) integrating the material into a coherent whole;
- Interactive lectures and seminars that challenge you to critically engage with the learning materials;
- Case studies designed to provide you with the opportunity to examine the theories, analyse situations, and develop and test relevant skills. (You will be split into groups where appropriate);
- Independent study is an essential element in order to make the most of the sessions;
- Essential and additional reading references that cover the topic area.

LECTURE/SEMINAR/LAB PROGRAMME

This module has been scheduled to run during Semester 1 and the lectures are scheduled as follows:

Week	Lecture Topic	Lecturer
1 1/10/2026	Scene Setting – Why study ethics and governance of digital systems?	SS
2	Introducing Ethical, Governance and Regulatory Contexts	SS

Week	Lecture Topic	Lecturer
8/10/2025		
3 15/10/2025	Ethics of AI and Large Language Models	SS
4 22/10/2025	Substantive Ethical Issues with Digital Technologies	SS
5 29/10/2025	Digital Governance in Organisations	AK
6	READING WEEK	
7 12/11/2025	Regulatory and Legal Frameworks	AK
8 19/11/2025	Participatory and Public Engagement	SS
9 26/11/2026	Responsible AI and Ethical Computing	SS
10 3/12/2025	Digital Technology Futures	AK
11 10/12/2025	Module Review and Exam Prep	SS
12 17/12/2025	Exam Support Q & A / Drop-in Session	SS

Seminars

Seminars will run each week in groups. Students will be allocated to groups by the module leader. Each seminar will explore issues related to the associated lecture topic. Preparatory reading will be required for these sessions. Seminars are scheduled as follows:

Week	Weekly Topic	Seminar Leaders
1 1/10/2025	No Seminar	-
2 8/10/2025	Self-Driving Vehicles Case	SS, AK, AP, FC
3 15/10/2025	Large Language Models in Higher Education	SS, AK, AP, FC
4 22/10/2025	NHS Federated Data Platform Case	SS, AK, AP, FC

Week	Weekly Topic	Seminar Leaders
5 29/10/2025	Sharing Economy & Platform Governance Case	SS, AK, AP, FC
6	READING WEEK	
7 12/11/2025	Facial Recognition Technology Case	SS, AK, AP, FC
8 19/11/2025	National/biometric Identity Card Case	
9 26/11/2026	Responsible AI Ethical Design Practice Case	SS, AK, AP, FC
10 3/12/2025	Automated Decision Systems in Public Services Case	SS, AK, AP, FC
11 10/12/2025	Algorithmic Decision-Making Systems (ADMS) Case	SS, AK, AP, FC

Please note: Week 6 in Semester 1 is a week during which there are no scheduled lectures, labs, seminars or tutorials. Weeks 15-17 are the Winter Examination Period. Week 18 is the first week of teaching after the Christmas break.

READING LIST

Core reading list

Articles in relevant journals will be recommended in the teaching sessions and will be required reading to supplement the lecture material. Specific readings for the seminars will be recommended each week and will be essential reading for effective participation in the discussions. The articles that are recommended will be shared on Brightspace.

Further readings:

- Kearns, M. and Roth, A. (2020). The Ethical Algorithm: The Science of Socially Aware Algorithm Design. Oxford University Press, New York
- Floridi, L. (2013). The ethics of information. Oxford University Press, UK
- Dubber, M. D., Pasquale, F., & Das, S. (Eds.). (2020). The Oxford handbook of ethics of AI. Oxford University Press

Supplementary Reading

Please note that there is a wealth of material, much of it freely available on the web or in the library. The above are suggestions, but you are encouraged to search for and make use of other sources.

It is important that you learn to become self-reliant and able to access and assimilate material for yourself. Many of the topics necessary for this project will not be covered by lectures.

Therefore, you will need to work through the exercises and guidance material provided in seminars, via Brightspace or from your own investigations.

ASSESSMENT

For all programmes except MSc DTSS, the assessment of the module will be through assessment block CS5805:

- 3-hour written examination, weighted at 100%
- The exam tests how well you can apply what you have learned to real situations. It does not test memorisation. You will analyse a case, evaluate governance frameworks, and argue for specific recommendations. The exam has two parts, and you need to pass the first before the second can be marked.

You can gather more information about this exam and the way the exam will be graded and assessed in the corresponding examination guide and grading schemes, which will be made available to you as detailed later on CS5805.

To prepare and do well in this exam, it is strongly recommended that you **attend All lectures and seminars**. The discussions and activities you do in the lectures and seminars will help you practise the skills you need for the exam. It is not enough to just be in the room; take part, share your ideas, and engage with the tasks. This is the best way to prepare.

EXPECTATIONS OF ARTIFICIAL INTELLIGENCE USE

The University has general guidance on using artificial intelligence in your studies.

The teaching and seminar activities require the students to critically engage with the weekly readings and lecture materials during lectures and seminars. Thus, the use of Generative AI tools (for example, ChatGPT, Claude or Perplexity) to substitute your own critical thinking is not encouraged. If you do not read and understand the learning materials yourself, it will be difficult to participate and learn from the sessions.

INCLUSIVITY STATEMENT

Seminars in this module offers the opportunity to collaborate with students from a variety of genders, ethnicities, cultural backgrounds, disabilities, neurodivergent profiles, and sexual orientations. Such power of diversity reflects the professional environment you will encounter in the industry and is a strength in achieving innovative and effective solutions. You are expected to value and respect different perspectives, foster a collaborative environment, and ensure that all members have the opportunity to contribute. This approach will support the development of your interpersonal, communication, and collaboration skills, which are essential for success in your future career.

ADDITIONAL VITAL INFORMATION

Information relating to all aspects of your studies, including procedures of how to inform us of problems you are facing with your studies, how to apply for an extension for your coursework submission, plagiarism and types of other academic misconduct, and other important matters, are available on the University's intranet. The Department assumes that you familiarise yourself with this information, so you will need to look at the relevant intranet pages carefully at various times throughout your studies. The Department also operates within the rules and regulations of the University more generally, and you should also look at what are known as 'Senate Regulations' on the University's web pages. These policies and

procedures might change from one academic year to another, and it is in your interest to review them on an on-going basis.